

## Lab innovation highlights

In today's fast-paced scientific environment, laboratories seek solutions that enhance both efficiency and accuracy. This is where the latest UV/Vis spectrophotometers come into play, offering a suite of features designed to meet these needs.

- **Versatile functionality.** These spectrophotometers are efficient and adaptable, serving both as classical spectrum-measuring devices and specialised colorimeters with automated colour calculations.
- **Seamless data management.** They support software integration for automated data export, streamlining data management and workflow.
- **Innovative features.** Features like barcode scanning and real-time path length monitoring enhance reliability and reduce errors.
- **Space-saving solution.** Compact yet multifunctional, they support various tests like titration, optimising space in labs.
- **User-friendly interface.** The simple interface includes guided workflows and a barcode reader for easy sample identification.
- **Effortless operation.** These devices automate measurements from start to finish, including path length identification and result generation, with a seven-day storage for blank measurements.

Embracing such cutting-edge technologies enables labs to stay ahead in a competitive field, ensuring high-quality results and streamlined operations. These spectrophotometers not only make complex tasks more manageable but also pave the way for future innovations in laboratory practices.

less likely to make mistakes. You will be able to quickly check the quality of your products because these machines give us results in seconds—it is all about fast and efficient analysis.

These spectrophotometers can analyse samples in an impressively short time. To ensure reliable testing, they're equipped with pre-programmed performance test methods. These methods help validate the instrument's performance, guaranteeing accurate and dependable results. Moreover, service technicians are available worldwide to perform validations using certified reference materials, providing professional support to maintain the instrument's accuracy.

When it comes to liquid samples, the simplicity of the process is even more enhanced as no sample preparation is necessary to check optical properties. This is crucial for us because any alteration to the sample, such as dilution or chemical treatment, could potentially distort the results. On the other hand, solid samples must be dissolved in an

appropriate solvent before measurement to ensure accurate readings. This step is essential for the correct analysis of solid materials, making sure that the results accurately reflect the true characteristics of the sample.

### Essential testing in many fields

UV/Vis spectroscopy is a versatile and widely used analytical technique, especially beneficial if you're involved in various industries. It offers a range of spectrophotometers, each designed to meet your specific needs, whether you're in pharmaceuticals, food and beverage, or environmental monitoring.

As someone deeply involved in quality control and research across various domains, including pharmaceuticals, food and beverage, and environmental monitoring, I find UV/Vis spectroscopy offers invaluable insights into the characteristics of samples. It's a crucial tool for ensuring product quality, studying chemical compositions, and monitoring environmental parameters, providing precise data

that guide my decisions and strategies.

However, it's important to remember that UV/Vis spectroscopy works best with samples that are clear and free of cloudiness. Choosing the right blank medium is crucial to get accurate results. This step is necessary to offset any effects the sample's environment might have on your measurements, leading to more precise and reliable analyses for your specific needs.

The exploration of ultraviolet/visible (UV/Vis) spectrophotometry within the realm of battery technology reveals its indispensable role in enhancing our understanding and management of battery health and efficiency. This analytical technique, with its profound ability to monitor the evolution of electrolyte solutions, stands as a cornerstone in the development of durable and efficient batteries. By offering detailed insights into electrolyte deterioration and impurity effects on battery performance, UV/Vis spectrophotometry not only aids in the rigorous assessment of battery health but also propels forward the research and development efforts aimed at creating more resilient battery technologies.

Its application transcends routine quality checks, emerging as a critical tool in the quest for innovation across various industries, including pharmaceuticals, food and beverage, and environmental monitoring. Through the lens of UV/Vis spectrophotometry, we gain a deeper understanding of the complexities of battery technology, paving the way for advancements that promise to redefine our approach to energy storage and utilisation. **EFY**

*Compiled by Nidhi Agarwal, Technology Journalist at EFY, this article features insights from Dr Fabian Müller, Product Manager for UV/Vis Spectrophotometers at METTLER TOLEDO, presented as a narrative of key discussion highlights*